Kubernetes

	Level	Duration
	Introduction/Advanced	3 days

Course Description

Audience and prerequisites

- This course is for DevOps developers who is intend or are currently working with K8S

Main Goals

- The main objective of this course is to get familiar with Kubernetes in a depth level
- Learn Containers and Kubernetes Fundamentals its architecture of (Control plane, components, worker nodes and more)
- Practice K8S Cluster Operations.
- HandsOn: deploying multiple resources & applications into Kubernetes.
- Become familiar with K8S YAML files & Kubernetes objects
- Understand Kubernetes core concepts & Objects.
- Use kubectl, the Kubernetes CLI, and become familiar with its commands and options
- Learn how to troubleshoot Kubernetes

Content

Session	**Content**
Introduction	
	🔹 What is K8s?

	🔸 Kubernetes Overview
	🔸 Introduction to Container Orchestration
	🔹 Kubernetes Architecture
	🔸 Kubernetes Core Concepts
Getting Started with Kubernetes	
	🔹 Labels & Selectors
	🔹 Kubernetes Clusters (tools for our HandsOn Labs)
	🔸 `minikube`
	🔸 `limactl`
	🔸 `Rancher`
	🔸 `k3d` / `microk8s` / `kind`
Kubernetes CLI	
	🔹 `kubectl`
	🔸 Commands
	🔸 JSONPath
Common Kubernetes Objects	
	🔹 Config files
	🔹 DaemonSet
	🔹 Deployments
	🔹 Jobs
	🔹 Pods
	🔹 ReplicaSets (+ Rolling update)
	🔹 StatefulSets
	🔹 Services
	🔹 CRD
Scaling Kubernetes / Auto-Scaling	

Manage Cluster content	
	🔹 Deploy multiple revisions
	🔹 Replica (sets, controller, etc)
	🔹 Update / Upgrade existing container
	🔹 Rollout
	🔹 init Containers
Health Checking	
	🔹 Liveness
	🔹 Readiness
	🔹 Resources
Kubernetes Networking	
	🔹 Services
	🔸 Ingress / Egress
	🔸 Network Policy
	🔸 ClusterIP, NodePort, and LoadBalancer
	🔹 Networking within a pod
	🔹 Pod-to-Pod Networking
	🔹 Ingress controllers
	🔹 DNS & Service Discovery
Stateful Applications in Kubernetes	
	🔹 Stateless versus Stateful
	🔹 Volumes
	🔸 Persistent volumes claims
	🔸 StorageClasses
	🔸 StatefulSets
Configuration	

	🔹 etcd
	🔹 Secrets
	🔹 ConfigMap
	🔹 Dynamic configuration
	🔹 Jobs, CronJobs
Security	
	🔹 RBAC
	🔹 SecurityContext
Auditing	
	🔹 logs
	🔹 events
	🔹 debugging
High Availability	
	🔹 Custom HPA
	🔹 Keda
	🔹 Backup cluster
Kubernetes API	
	🔹 Admission Controls
	🔹 hooks
	🔹 api/metrics
Additional Content	
	🔹 Kustomization
	🔹 CRD
	🔹 HELM
	🔸 What is Helm
	🔸 Working with Helm

	🔸 Build Helm package
	🔹 Tools: (Most of this section is hands on)
	🔸 Service Mesh
	🔸 Istio / Linkerd
	🔸 Keda
	🔸 Lens
	🔸 Prometheus
	🔸 Grafana
	🔸 ArgoCD

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